Dynamic Models

Professor: Juan David Bastidas Rodríguez

Reference book for the presentation: **Feedback Control of Dynamic Systems, 8th Edition, Franklin • Powell • Emami-Naeini, Pearson, 2019, ISBN: 978-0-13-468571-7**

Content

1. Introduction to dynamic models
2. Models of Electric Circuits
3. Dynamics of Mechanical Systems
4. Heat and Fluid-Flow Models
5. Models of Electromechanical Systems (optional)

1. Introduction to dynamic models

The **first step** to obtain a closed–loop control system is to develop a **mathematical description** (or dynamic model or mathematical model) of the **process to be controlled**

- **Model:** a set of **differential equations** that describe the **dynamic behavior** of the process
  - Based on physics (we will focus on this) => verified with experiments
  - Based on input-output response vs time (system identification)

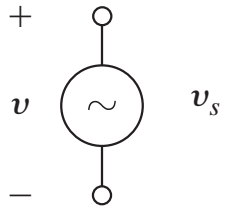
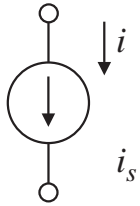
2. Dynamics of Electric Systems

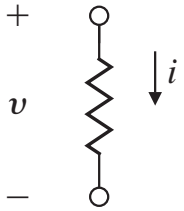
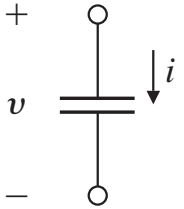
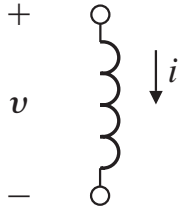
Electric Circuits:

- Some applications require analog controllers because:
  - Faster response
  - Lower cost
- Used for electromechanical energy conversion (e.g. electric motors, generators, and electrical sensors)
- Analog filters required for controllers (digital and analog)
- Operational amplifiers (op-amp) are important and used in:
  - Analog controllers (compensators)
  - Analog filters

2. Dynamics of Electric Systems

Electric Circuits:

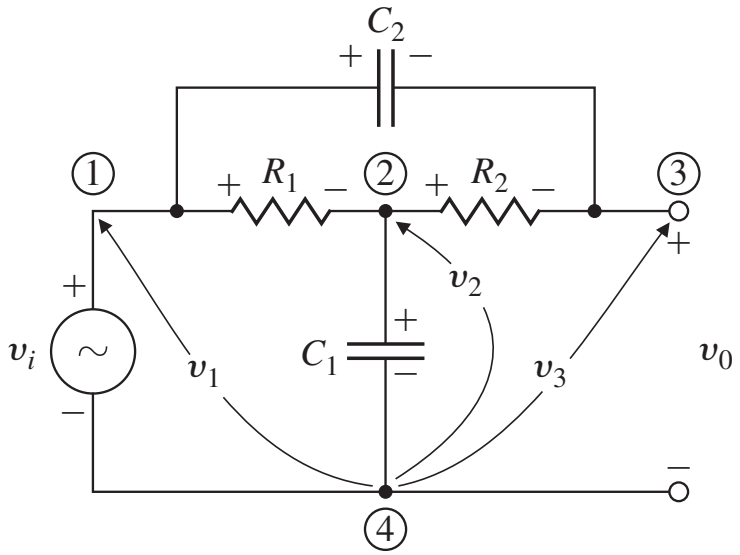
	Symbol	Equation
Voltage source		$v = v_s$
Current source		$i = i_s$

	Symbol	Equation
Resistor		$v = Ri$
Capacitor		$i = C\frac{dv}{dt}$
Inductor		$v = L\frac{di}{dt}$

2. Dynamics of Electric Systems

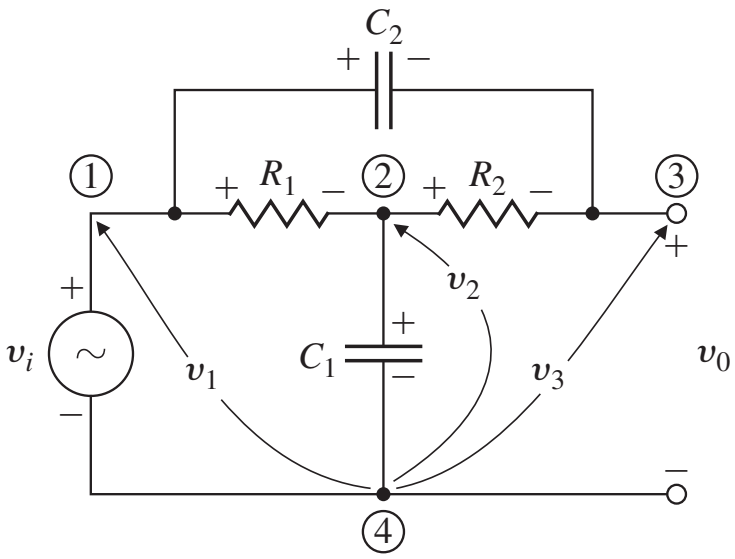
Electric Circuits: Bridged Tee Circuit

- Passive circuit to reduce signal power (attenuate) without significantly altering the circuit's impedance
- Useful in various applications like RF and audio systems



2. Dynamics of Electric Systems

Electric Circuits: Bridged Tee Circuit



$$v_1 = v_i.$$

$$-\frac{v_1 - v_2}{R_1} + \frac{v_2 - v_3}{R_2} + C_1 \frac{dv_2}{dt} = 0,$$

$$\frac{v_3 - v_2}{R_2} + C_2 \frac{d(v_3 - v_1)}{dt} = 0.$$

Replace d()/dt by "s", v_i by Vi(s), v_o by Vo(s) to obtain the transfer function Vo(s)/Vi(s)
$R_1 = R_2 = 1.5\ k\Omega$, $C_1 = C_2 = 0.1\ \mu F$

- Find the forced response for DC and AC excitation
- Find the the transient response
- Find the complete response
- Find the poles and zeros

3. Dynamics of Mechanical Systems

Translational motion: Newton's law

$$\mathbf{F} = m\mathbf{a},$$

$\mathbf{F}$ = the vector sum of all forces applied to each body in a system, newtons (N),

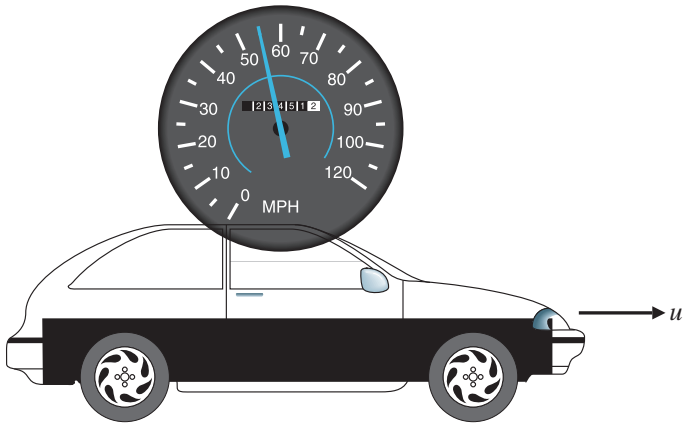
$\mathbf{a}$ = the vector acceleration of each body with respect to an **inertial reference frame** (that is, one that is neither accelerating nor rotating with respect to the stars); often called **inertial acceleration**, m/sec^2,

m = mass of the body, kg.

3. Dynamics of Mechanical Systems

Translational motion: A Simple System; Cruise Control Model

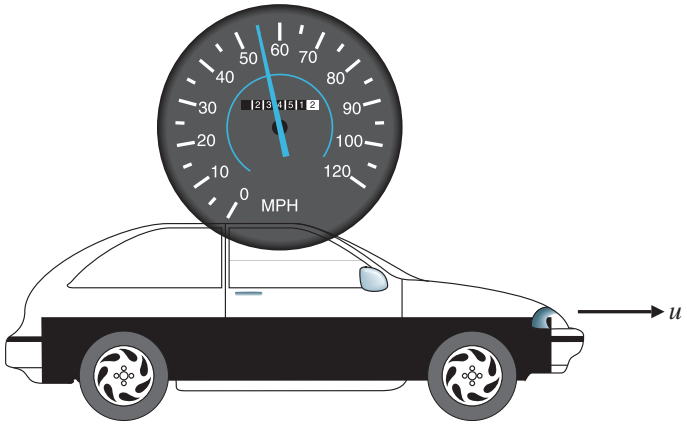
1. Write the equations of motion
2. Use Matlab to find the response of the velocity
    1. u = 0 at time t = 0 to a
    2. constant u = 500 N t > 0
    3. m = 1000 kg
    4. Viscous drag coeff. b = 50 N·sec/m.

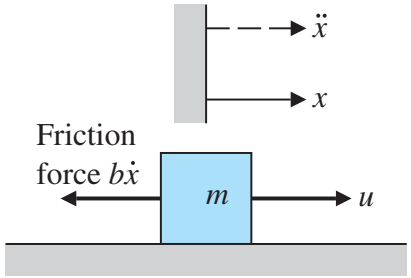


3. Dynamics of Mechanical Systems

Translational motion: A Simple System; Cruise Control Model

1. Equations of motion





$$u - b\dot{x} = m\ddot{x},$$

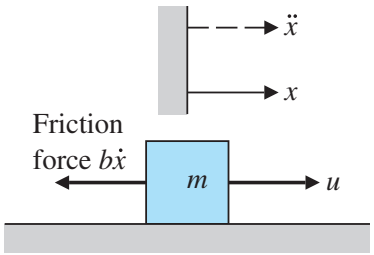
$$\ddot{x} + \frac{b}{m}\dot{x} = \frac{u}{m}.$$

$$\dot{v} + \frac{b}{m}v = \frac{u}{m}.$$

3. Dynamics of Mechanical Systems

Translational motion: A Simple System; Cruise Control Model

1. Equations of motion: interest variable => speed (v)



$$\dot{v} + \frac{b}{m}v = \frac{u}{m}.$$

$$v = V_o e^{st}$$

$$\dot{v} = sV_o e^{st}$$

$$u = U_o e^{st}$$

$$\left(s + \frac{b}{m}\right) V_o e^{st} = \frac{1}{m} U_o e^{st}$$

$$\frac{V_o}{U_o} = \frac{\frac{1}{m}}{s + \frac{b}{m}}$$

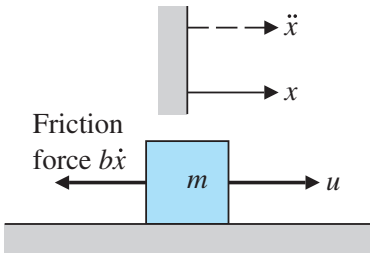
Transfer function

$$\frac{V(s)}{U(s)} = \frac{\frac{1}{m}}{s + \frac{b}{m}}$$

3. Dynamics of Mechanical Systems

Translational motion: A Simple System; Cruise Control Model

2. Time response



Transfer function

$$\frac{V(s)}{U(s)} = \frac{\frac{1}{m}}{s + \frac{b}{m}}$$

Matlab:

s=tf('s'); % sets up the mode to define the transfer function

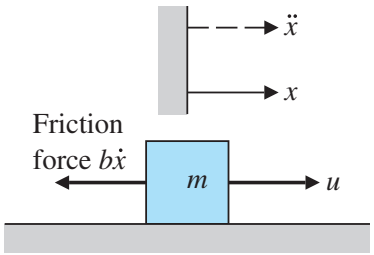
sys = (1/1000)/(s + 50/1000); % defines the transfer function from Eq. (2.7) with the numbers filled in.

step(500*sys); % plots the step response for u = 500.

3. Dynamics of Mechanical Systems

Translational motion: A Simple System; Cruise Control Model

2. Time response

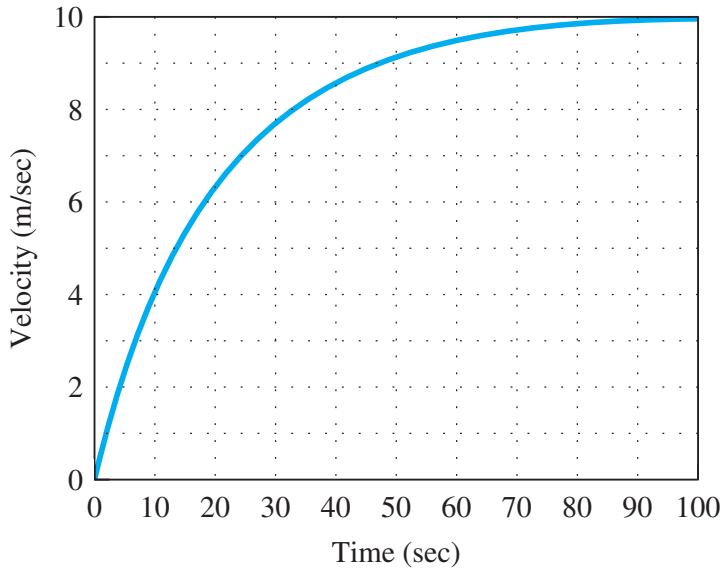


Transfer function

$$\frac{V(s)}{U(s)} = \frac{\frac{1}{m}}{s + \frac{b}{m}}$$

Matlab:

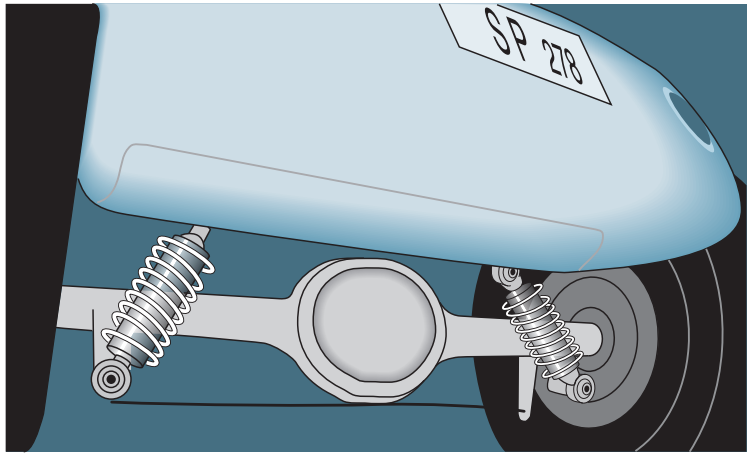
```
s=tf('s');
sys = (1/1000)/(s + 50/1000);
step(500*sys);
```



3. Dynamics of Mechanical Systems

Translational motion: A Two-Mass System: Suspension Model

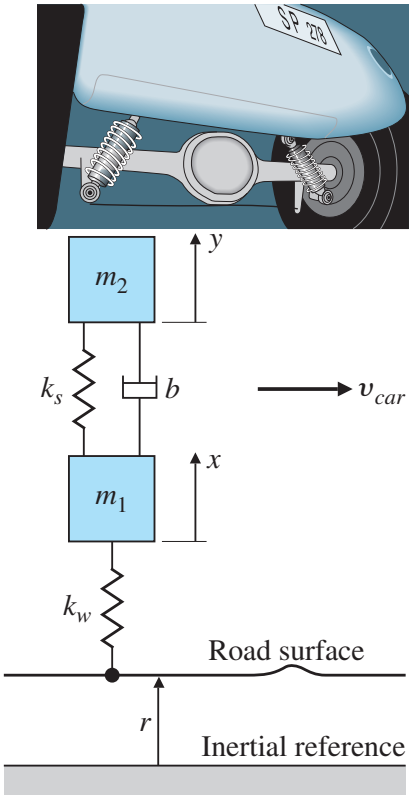
- Car mass 1580 kg
- 1/4 of the mass on each wheel
- Wheel mass 20 kg
- Suspension k_s: Applying a known force (weight of a person) and measuring the deflection (distance) k_s = 130,000 N/m
- Wheel k_w: Applying a known force (weight of a person) and measuring the deflection (distance) k_w = 1,000,000 N/m
- Assuming a damping coefficient of 0.7 from the time response (ζ= 0.7) it is possible to obtain b = 9800 N·sec/m



3. Dynamics of Mechanical Systems

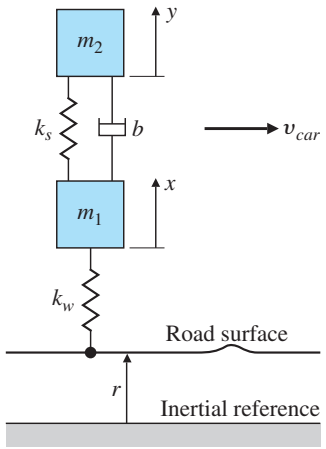
Translational motion: A Two-Mass System: Suspension Model

- x and y: the displacements of the masses from their equilibrium
  - Gravity doesn't need to be included => only affects the equilibrium
  - Gravity considered in the equilibrium
- Shock absorber force= b(y' – x') (relative displacement of m_2 w.r.t m_1)

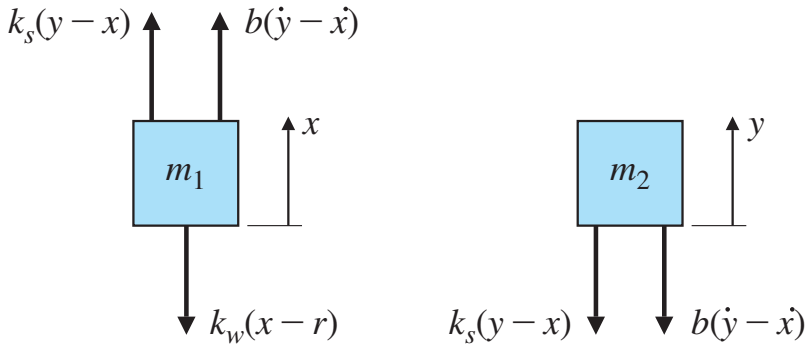


3. Dynamics of Mechanical Systems

Translational motion: A Two-Mass System: Suspension Model



Verify the signs !!!
Consider this is a stable system

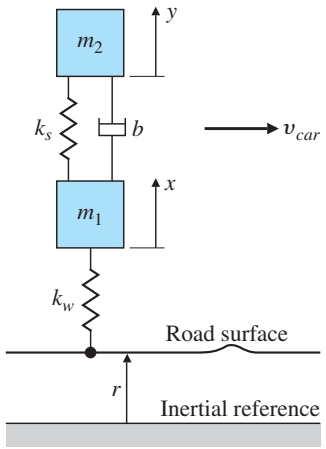


$$b(\dot{y}-\dot{x})+k_s(y-x)-k_w(x-r)=m_1\ddot{x},$$
$$-k_s(y-x)-b(\dot{y}-\dot{x})=m_2\ddot{y}.$$

$$\ddot{x}+\frac{b}{m_1}(\dot{x}-\dot{y})+\frac{k_s}{m_1}(x-y)+\frac{k_w}{m_1}x=\frac{k_w}{m_1}r,$$
$$\ddot{y}+\frac{b}{m_2}(\dot{y}-\dot{x})+\frac{k_s}{m_2}(y-x)=0.$$

3. Dynamics of Mechanical Systems

Translational motion: A Two-Mass System: Suspension Model



$$\ddot{x} + \frac{b}{m_1}(\dot{x} - \dot{y}) + \frac{k_s}{m_1}(x - y) + \frac{k_w}{m_1}x = \frac{k_w}{m_1}r,$$

$$\ddot{y} + \frac{b}{m_2}(\dot{y} - \dot{x}) + \frac{k_s}{m_2}(y - x) = 0.$$

Substituting s for d/dt

$$s^2X(s) + s\frac{b}{m_1}(X(s) - Y(s)) + \frac{k_s}{m_1}(X(s) - Y(s)) + \frac{k_w}{m_1}X(s) = \frac{k_w}{m_1}R(s)$$

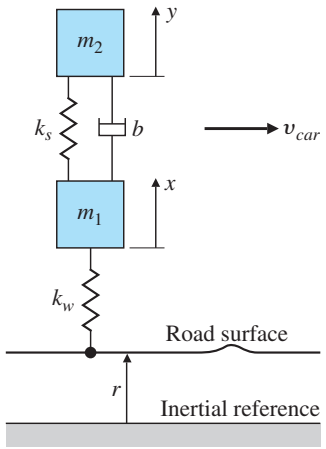
$$s^2Y(s) + s\frac{b}{m_2}(Y(s) - X(s)) + \frac{k_s}{m_2}(Y(s) - X(s)) = 0,$$

In matrix form

$$\begin{bmatrix} s^2 + s\frac{b}{m_1} + \frac{k_s}{m_1} + \frac{k_w}{m_1} & -s\frac{b}{m_1} - \frac{k_s}{m_1} \\ -s\frac{b}{m_2} - \frac{k_s}{m_2} & s^2 + s\frac{b}{m_2} + \frac{k_s}{m_2} \end{bmatrix} \begin{bmatrix} X(s) \\ Y(s) \end{bmatrix} + \begin{bmatrix} \frac{k_w}{m_1} \\ 0 \end{bmatrix} R(s)$$

3. Dynamics of Mechanical Systems

Translational motion: A Two-Mass System: Suspension Model



$$\begin{bmatrix} s^2 + s\frac{b}{m_1} + \frac{k_s}{m_1} + \frac{k_w}{m_1} & -s\frac{b}{m_1} - \frac{k_s}{m_1} \\ -s\frac{b}{m_2} - \frac{k_s}{m_2} & s^2 + s\frac{b}{m_2} + \frac{k_s}{m_2} \end{bmatrix} \begin{bmatrix} X(s) \\ Y(s) \end{bmatrix} + \begin{bmatrix} \frac{k_w}{m_1} \\ 0 \end{bmatrix} R(s)$$

By applying Cramer's Rule for Y(s)

$$\frac{Y(s)}{R(s)} = \frac{\frac{k_w b}{m_1 m_2}\left(s + \frac{k_s}{b}\right)}{s^4 + \left(\frac{b}{m_1} + \frac{b}{m_2}\right)s^3 + \left(\frac{k_s}{m_1} + \frac{k_s}{m_2} + \frac{k_w}{m_1}\right)s^2 + \left(\frac{k_w b}{m_1 m_2}\right)s + \frac{k_w k_s}{m_1 m_2}}$$

$m_1 = 20$ kg, $m_2 = (1580 - 4*20)/4 = 375$ kg, $k_s = 130$ kN/m, $k_w = 1$ MN/m, $b = 9.8$ kN·sec/m

$$\frac{Y(s)}{R(s)} = \frac{1.31e06(s + 13.3)}{s^4 + (516.1)s^3 + (5.685e04)s^2 + (1.307e06)s + 1.733e07}$$

3. Dynamics of Mechanical Systems

Rotational motion: Newton's law for one-dimensional rotational system

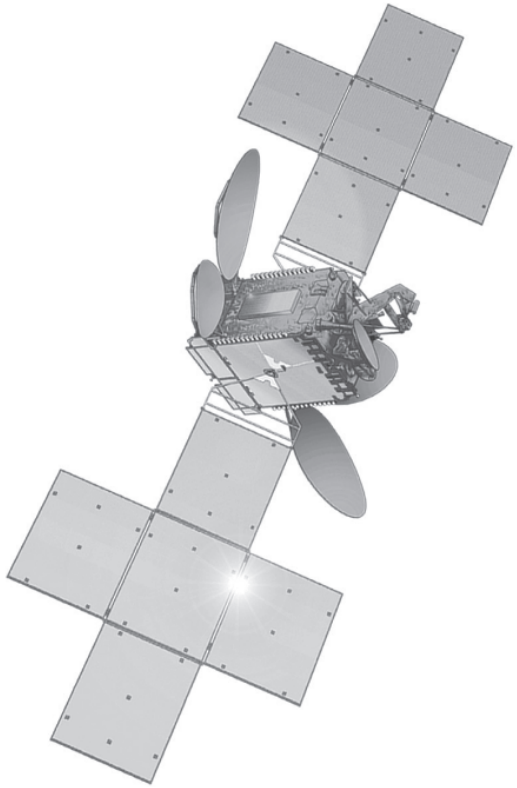
$$M = I\alpha$$

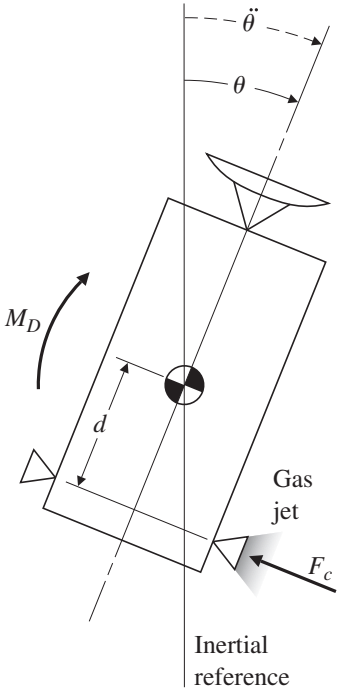
$M =$ the sum of all external moments about the center of mass of a body, $\mathrm{N} \cdot \mathrm{m}$,

$I =$ the body's mass moment of inertia about its center of mass, $\mathrm{kg} \cdot \mathrm{m}^2$,

$\alpha =$ the angular acceleration of the body, $\mathrm{rad/sec}^2$.

3. Dynamics of Mechanical Systems



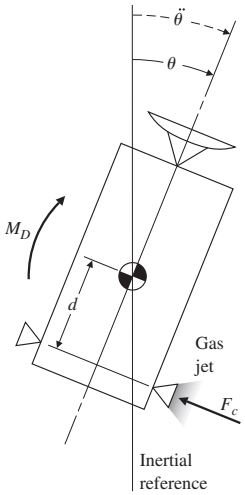


Rotational motion: Satellite Attitude Control Model

- Considering 1 axis first
- θ: satellite orientation
- Inertial reference (no angular acceleration)
- F_c: force produced by the jet
- M_D: torque produced by solar pressure difference in solar panels of both sides

3. Dynamics of Mechanical Systems

Rotational motion:
Satellite Attitude Control Model



$$F_c d + M_D = I\ddot{\theta}.$$

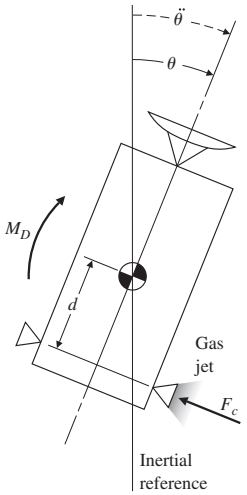
To obtain θ => integrate the sum of the input torques twice (**double-integrator plant**)

Substituting U = F_{cd} + M_D and s for d/dt

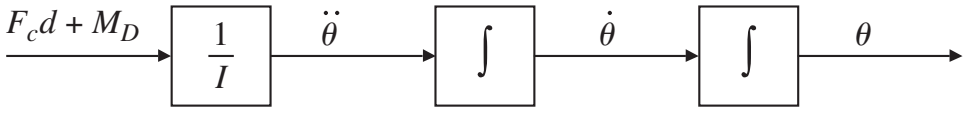
$$\frac{\Theta(s)}{U(s)} = \frac{1}{I}\frac{1}{s^2}$$

3. Dynamics of Mechanical Systems

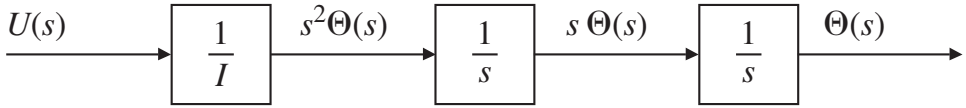
Rotational motion:
Satellite Attitude Control Model



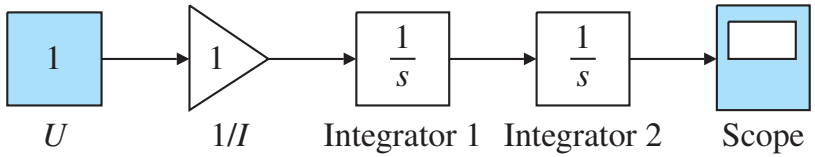
$$F_c d + M_D = I\ddot{\theta}.$$



$$\frac{\Theta(s)}{U(s)} = \frac{1}{I}\frac{1}{s^2}$$



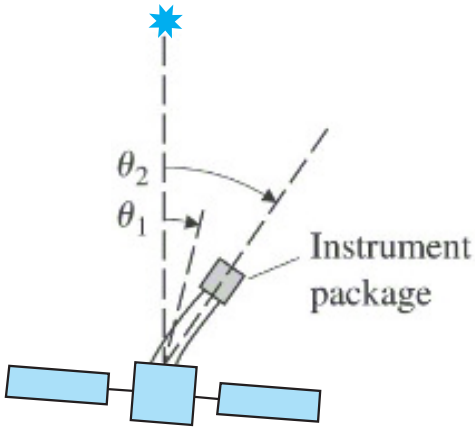
Simulink block diagram of the **double-integrator** plant

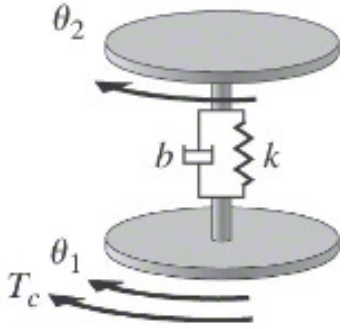


3. Dynamics of Mechanical Systems

Rotational motion: Flexibility: Flexible Satellite Attitude Control

- Difference between the satellite attitude sensor or orientation sensor (θ2) and the body of the satellite (θ1) where the actuators are placed
- **Typical** of many situations in which the sensor and actuator may or may not be placed in the same location in a flexible body
- What is the motion of the instrument package to a control torque applied to the body of the satellite?

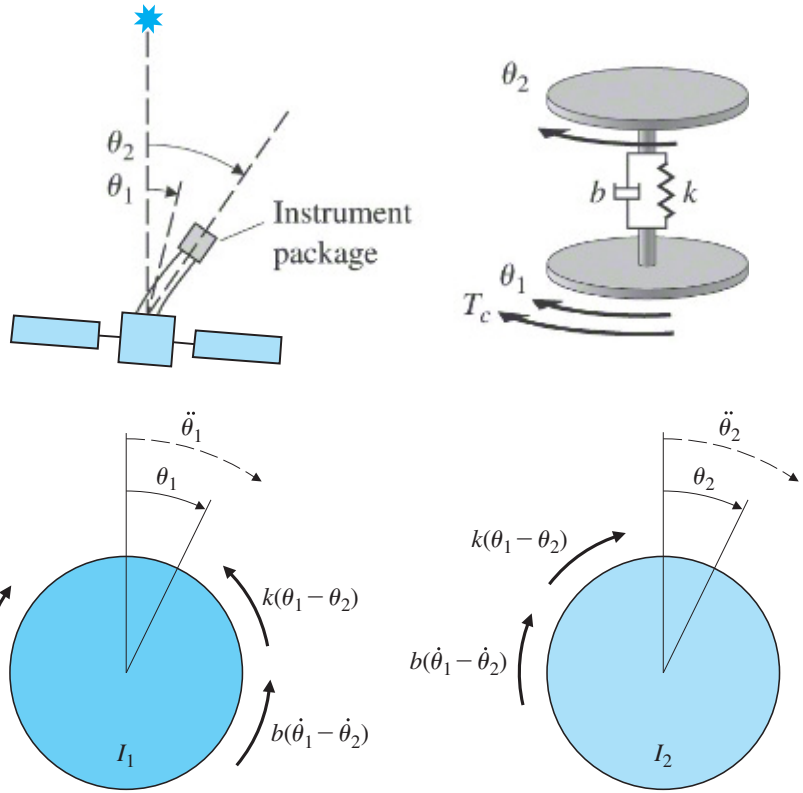




3. Dynamics of Mechanical Systems

Rotational motion:
Flexibility: Flexible Satellite Attitude Control

- Similar to the example of the car shock absorbers but with moments (torques)
- T_c: moment produced by the actuators (jets)
- k: spring constant
- b: friction constant

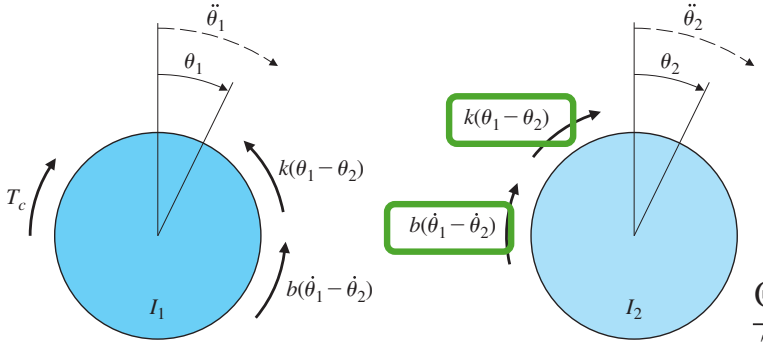


3. Dynamics of Mechanical Systems

Rotational motion:
Flexibility: Flexible Satellite Attitude Control

Pay attention to the signs

$$I_1\ddot{\theta}_1 + b(\dot{\theta}_1 - \dot{\theta}_2) + k(\theta_1 - \theta_2) = T_c$$
$$I_2\ddot{\theta}_2 + b(\dot{\theta}_2 - \dot{\theta}_1) + k(\theta_2 - \theta_1) = 0.$$



Ignoring the damping b for simplicity, and substituting s for d/dt

$$(I_1 s^2 + k)\Theta_1(s) - k\Theta_2(s) = T_c$$
$$-k\Theta_1(s) + (I_2 s^2 + k)\Theta_2(s) = 0.$$

Using Cramer's Rule

$$\frac{\Theta_1(s)}{T_c(s)} = \frac{I_2 s^2 + k}{I_1 I_2 s^2\left(s^2 + \frac{k}{I_1} + \frac{k}{I_2}\right)}. \qquad \frac{\Theta_2(s)}{T_c(s)} = \frac{k}{I_1 I_2 s^2\left(s^2 + \frac{k}{I_1} + \frac{k}{I_2}\right)}.$$

4. Models of Electromechanical systems

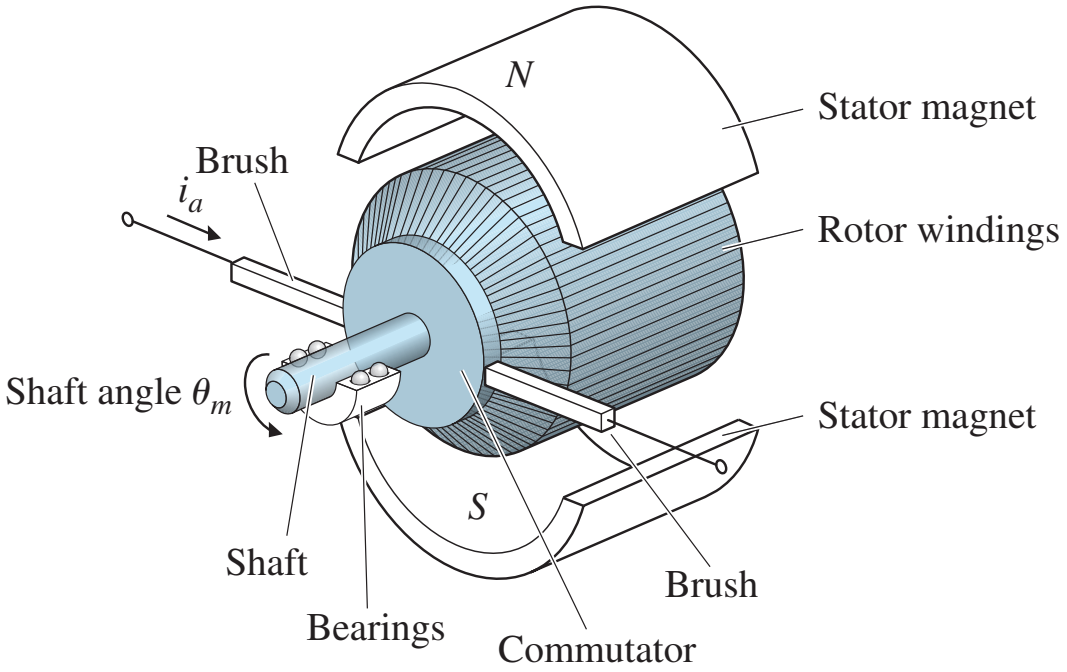
- Electric current and magnetic fields interact
- A current flowing through a cable generates a magnetic field
- The electrons flowing through a cable of distance “l” with a current “i” inside of a perpendicular magnetic field “B” experience a force F

$$F = Bli \text{ N}.$$

  - This is the key force for the electric motors’ operation

4. Models of Electromechanical systems

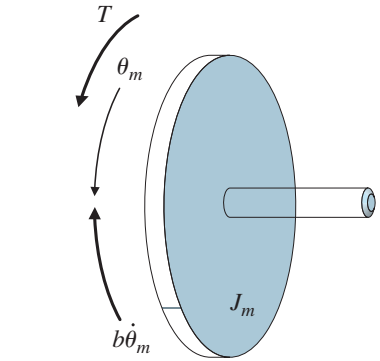
DC Motor:

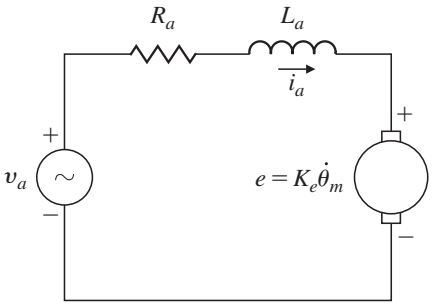


Explanation animation

4. Models of Electromechanical systems

DC Motor:





$K_t = K_e$ but usually given in different units

$$T = K_t\, i_a,$$
$$e = K_e\, \dot{\theta}_m.$$

T: torque on the rotor
i_a: armature current
e: back emf voltage
$\dot{\theta}_m$: shaft's rotational velocity

$$J_m\ddot{\theta}_m + b\dot{\theta}_m = K_t i_a$$

From the free-body diagram for the rotor

$$L_a\frac{di_a}{dt} + R_a i_a = v_a - K_e\dot{\theta}_m$$

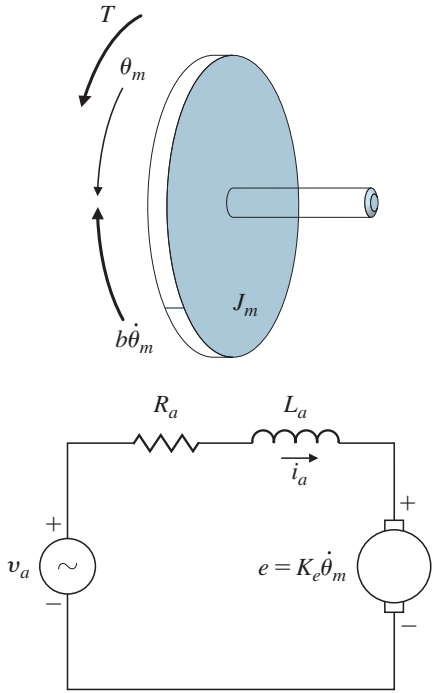
From the electric circuit

$$\frac{\Theta_m(s)}{V_a(s)} = \frac{K_t}{s\left[(J_m s + b)(L_a s + R_a) + K_t K_e\right]}$$

d/dt substituted for s and organizing

4. Models of Electromechanical systems

DC Motor:



In many cases the relative effect of the inductance is negligible compared with the mechanical motion

$$J_m\ddot{\theta}_m + \left(b + \frac{K_t K_e}{R_a}\right)\dot{\theta}_m = \frac{K_t}{R_a}v_a.$$

$$\frac{\Theta_m(s)}{V_a(s)} = \frac{\frac{K_t}{R_a}}{J_m s^2 + \left(b + \frac{K_t K_e}{R_a}\right)s} = \frac{K}{s(\tau s + 1)}$$

$$K = \frac{K_t}{bR_a + K_t K_e} \qquad \tau = \frac{R_a J_m}{bR_a + K_t K_e}$$

It is possible to obtain the TF of the speed with respect to the voltage

$$\frac{\Omega(s)}{V_a(s)} = s\frac{\Theta_m(s)}{V_a(s)} = \frac{K}{\tau s + 1}$$

5. Heat and Fluid-Flow models

- Using thermodynamics, heat transfer, and fluid dynamics to generate models
- Require experiments to determine the parameters
- Each topic is complex, i.e. there are complete textbooks for each one
- Models required in different industrial process
  - Tanks
  - Heat exchangers
  - Temperature control of liquids
  - Many other processes

5. Heat and Fluid-Flow models

Heat Flow:

- Temperature regulation for portions of the system
  - Flow and storage of heat energy
- Heat energy flows through substances at a rate proportional to the temperature difference across the substance

Analog to Ohm's law

$$q = \frac{1}{R}(T_1 - T_2)$$

q = heat-energy flow, joules per second (J/sec),
R = thermal resistance, (°C·sec)/J
T = temperature, °C.

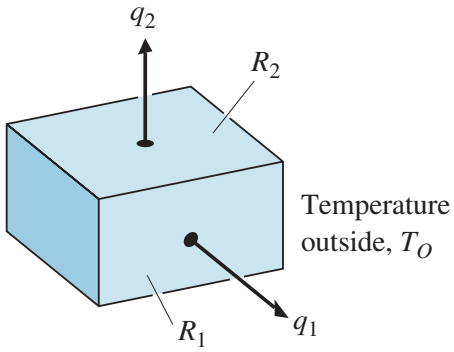
Analog to electrical capacitor's equation

$$\dot{T} = \frac{1}{C}q$$

C is the thermal capacity.

5. Heat and Fluid-Flow models

Heat flow for a room: what is the room temperature?



$$\dot{T}_I = \frac{1}{C_I}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)(T_O - T_I)$$

C_I = thermal capacity of air within the room,
T_O = temperature outside,
T_I = temperature inside,
R_2 = thermal resistance of the room ceiling,
R_1 = thermal resistance of the room wall.

C calculated from c_v (specific heat at constant volume) and m (mass of the substance)

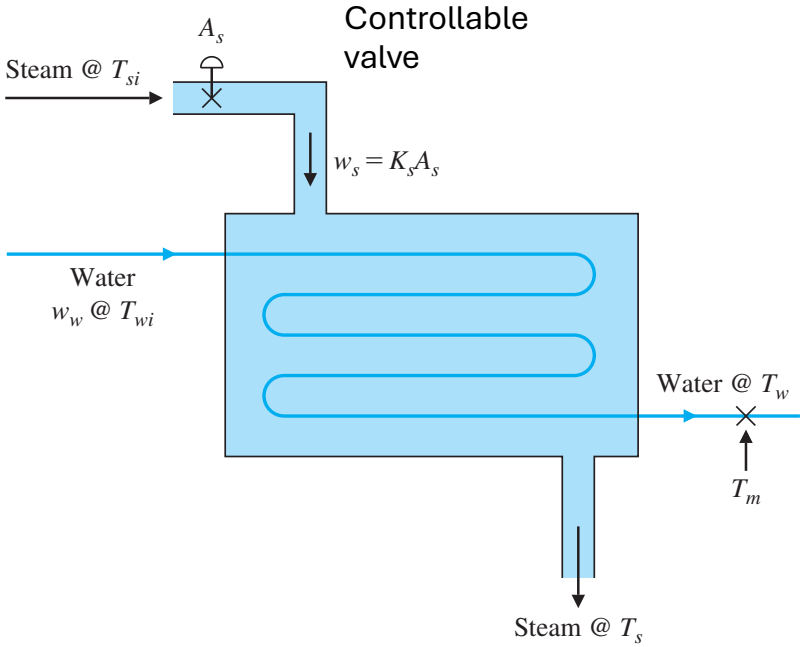
$$C = mc_v,$$

R calculated from k (thermal conductivity) and A (cross-sectional area) and l (the length of the heat-flow path)

$$\frac{1}{R} = \frac{kA}{l}$$

5. Heat and Fluid-Flow models

Heat exchanger: input -> A_s (valve area), output -> T_m (temp.)



Heat flow into the chamber q_{in}

$$q_{in} = w_s c_{vs}(T_{si} - T_s)$$

$w_s = K_s A_s$, mass flow rate of the steam,

A_s = area of the steam inlet valve,

K_s = flow coefficient of the inlet valve,

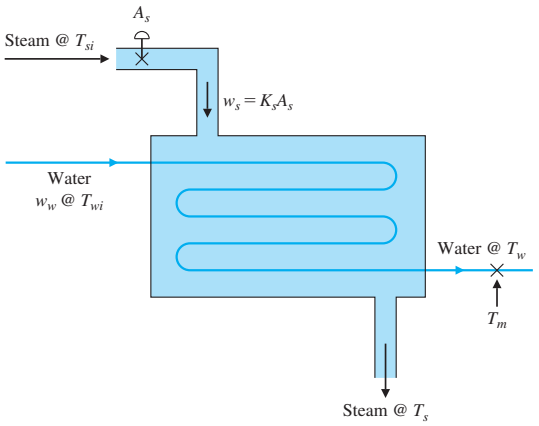
c_{vs} = specific heat of the steam,

T_{si} = temperature of the inflow steam,

T_s = temperature of the outflow steam.

5. Heat and Fluid-Flow models

Heat exchanger:
input -> A_s (valve area), output -> T_m (temp.)



$$C_s \dot{T}_s = A_s K_s c_{vs}(T_{si} - T_s) - \frac{1}{R}(T_s - T_w)$$

$w_s = K_s A_s$, mass flow rate of the steam,

A_s = area of the steam inlet valve,

K_s = flow coefficient of the inlet valve,

c_{vs} = specific heat of the steam,

T_{si} = temperature of the inflow steam,

T_s = temperature of the outflow steam.

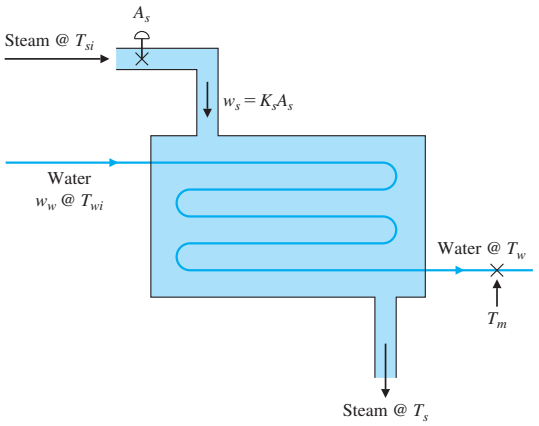
$$C_w \dot{T}_w = w_w c_{cw}(T_{wi} - T_w) + \frac{1}{R}(T_s - T_w)$$

$C_s = m_s c_{vs}$ is the thermal capacity of the steam in the chamber with mass m_s,

R = the thermal resistance of the heat flow averaged over the entire exchanger.

5. Heat and Fluid-Flow models

Heat exchanger: input -> A_s (valve area), output -> T_m (temp.)



Time delay (t_d) in the measurement

$$T_m(t) = T_w(t - t_d)$$

Nonlinear: T_s is multiplied by the control input A_s

1. $T_{si} - T_s$: assumed constant and named ΔT_s
2. T_w and T_s measured as deviation from Twi, i.e. $T_{w1} = T_w - T_{wi}$, and $T_{s1} = T_s - T_{wi}$ and T_{wi} constant at the operating point

$$C_s \dot{T}_{s1} = -\frac{1}{R} T_{s1} + \frac{1}{R} T_{w1} + K_s c_{vs} \Delta T_s A_s,$$

$$C_w \dot{T}_{w1} = -\left(\frac{1}{R} + w_w c_{vw}\right) T_{w1} + \frac{1}{R} T_{s1}$$

$$T_m = T_{w1}(t - t_d).$$

$$\frac{T_m(s)}{A_s(s)} = \frac{K e^{-t_d s}}{(\tau_1 s + 1)(\tau_2 s + 1)}.$$

5. Heat and Fluid-Flow models

Incompressible Fluid Flow:

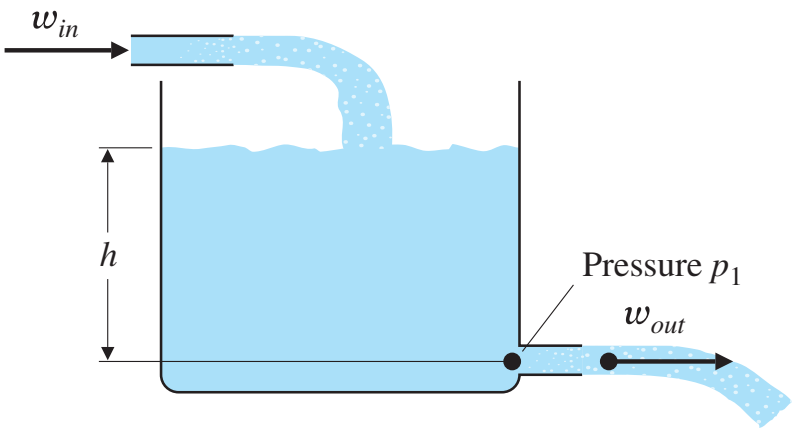
- The physical relations governing fluid flow are:
  - Continuity (conservation of matter)
  - Force equilibrium
  - Resistance

$$\dot{m} = w_{in} - w_{out},$$

$$m = \text{fluid mass within a prescribed portion of the system,}$$
$$w_{in} = \text{mass flow rate into the prescribed portion of the system,}$$
$$w_{out} = \text{mass flow rate out of the prescribed portion of the system.}$$

5. Heat and Fluid-Flow models

Incompressible Fluid Flow: Water Tank Height



$$\dot{h} = \frac{1}{A\rho}(w_{in} - w_{out})$$

A = area of the tank,

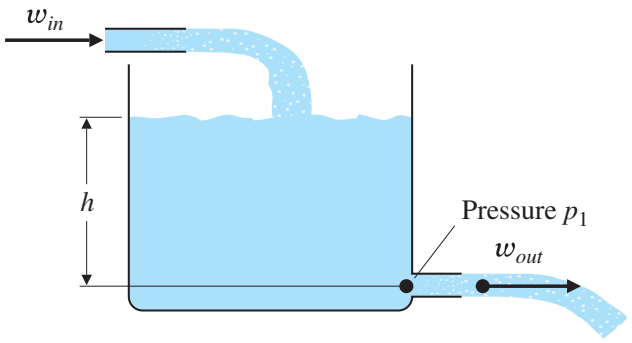
ρ = density of water,

$h = m/A\rho$ = height of water,

m = mass of water in the tank.

5. Heat and Fluid-Flow models

Incompressible Fluid Flow: Linearization of Water Tank Height and Outflow



The general form of the resistance

$$w = \frac{1}{R}(p_1 - p_2)^{1/\alpha}$$

w = mass flow rate,

p_1, p_2 = pressures at ends of the path through which flow is occurring,

R, α = constants whose values depend on the type of restriction.

$$Q = \frac{1}{\rho R}(p_1 - p_2)^{1/\alpha}$$

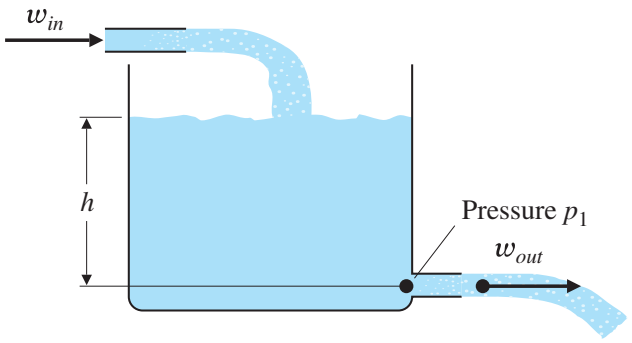
Q = volume flow rate, where $Q = w/\rho$,

ρ = fluid density.

α varies from 1 (laminar flow) to 2 (turbulent flow)

5. Heat and Fluid-Flow models

Incompressible Fluid Flow: Linearization of Water Tank Height and Outflow



$$w_{out} = \frac{1}{R}(p_1 - p_a)^{1/2}.$$

$p_1 = \rho g h + p_a$, the hydrostatic pressure,

p_a = ambient pressure outside the restriction.

Replacing in the initial tank equation

$$\dot{h} = \frac{1}{A\rho}\left(w_{in} - \frac{1}{R}\sqrt{p_1 - p_a}\right)$$

Nonlinear equation

Linearize around an operating point

$$p_o = \rho g h_o + p_a$$

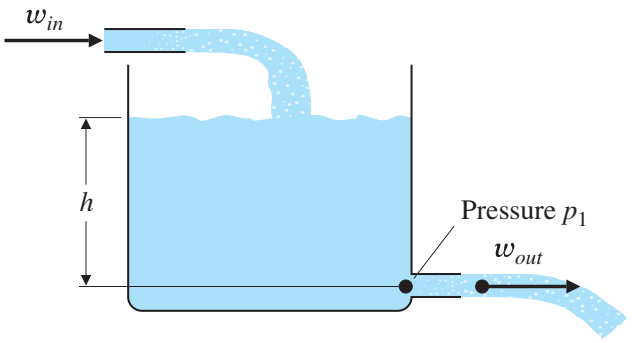
Using (valid when ε << 1)

$$(1+\varepsilon)^{\beta} \cong 1 + \beta\varepsilon,$$

Assuming $\alpha = 2$ (turbulent flow)

5. Heat and Fluid-Flow models

Incompressible Fluid Flow: Linearization of Water Tank Height and Outflow



$$w_{out} = \frac{\sqrt{p_o - p_a}}{R}\left(1 + \frac{\Delta p}{p_o - p_a}\right)^{1/2}$$

Rewriting w_{out} and replacing

$$\cong \frac{\sqrt{p_o - p_a}}{R}\left(1 + \frac{1}{2}\frac{\Delta p}{p_o - p_a}\right).$$

$$\Delta\dot{h} = \frac{1}{A\rho}\left[w_{in} - \frac{\sqrt{p_o - p_a}}{R}\left(1 + \frac{1}{2}\frac{\Delta p}{p_o - p_a}\right)\right]$$

Rewriting Δp in terms of Δh $\quad \Delta p = \rho g \Delta h,$

What happens if $\Delta h = 0$ and $w_{in} = 0$? $\quad \Delta\dot{h} = -\frac{g}{2AR\sqrt{p_o - p_a}}\Delta h + \frac{w_{in}}{A\rho} - \frac{\sqrt{p_o - p_a}}{\rho AR},$

Summary

Model can be obtained from:

- Application of physical laws (differential equations)
- From experimental data
- The basic equations are shown in the table

Key Equations for Dynamic Models

System	Important Laws or Relationships	Associated Equations
Mechanical	Translational motion (Newton's law)	$F = m\mathbf{a}$
	Rotational motion	$M = I\alpha$
Electrical	Operational amplifier	
Electromechanical	Law of motors	$F = Bli$
	Law of generators	$e = Blv$
	Torque developed in a rotor	$T = K_t i_a$
Back emf	Voltage generated as a result of rotation of a rotor	$e = K_e \dot{\theta}_m$
Gears	Effective inertia	$J_{eq} = J_2 + J_1 n^2$
Heat flow	Heat-energy flow	$q = {}^1/_R(T_1 - T_2)$
	Temperature as a function of heat-energy flow	$\dot{T} = \frac{1}{C} q$
	Specific heat	$C = mc_v$
Fluid flow	Continuity relation (conservation of matter)	$\dot{m} = w_{in} - w_{out}$
	Force of a fluid acting on a piston	$f = pA$
	Effect of resistance to fluid flow	$w = {}^1/_R(p_1 - p_2)^{1/\alpha}$